HACKWISE X

DATA CENTERIC IMAGE CLASSIFICATION

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ENDURANCE

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This project implements a binary image classifier to distinguish between chihuahuas and muffins using a data-centric AI approach. Instead of modifying the model architecture, we improved model performance by strategically labelling samples from an unlabelled dataset using embeddings provided by the 3LC platform. Through multiple iterations of labelling and retraining, the dataset quality improved, resulting in higher classification accuracy on the Kaggle test set.

DATASET DESCRIPTION:

- **Train:** 100 labelled (50 chihuahua, 50 muffin) + 3,579 unlabelled (undefined). Use 3LC to label part of the unlabelled set and curate which samples to train on.
- **Val:** 1,000 images (500 per class, balanced); use for monitoring during development. Gives stable feedback so the model generalizes well.
- **Test:** 1,184 images, hidden labels; you only get images. Predictions go into submission.csv and are scored on Kaggle (see **Evaluation** tab for format).

METHODOLOGY

- In this project, a **ResNet-18 convolutional neural network** was used to classify images as **chihuahua (0)** or **muffin (1)**. As per the competition rules, the model architecture was fixed and trained **from scratch using PyTorch**, without using pretrained weights or external data.
- Initially, the model was trained using the **100 labelled images** provided in the dataset (50 chihuahua and 50 muffin). The remaining images were unlabelled and excluded from training using **sample weights**.
- After training, **embeddings and predictions** generated by the model were visualized in the **3LC dashboard**. These embeddings helped identify clusters of visually similar images. Based on this visualization, selected unlabelled samples were labelled using a **data-centric approach**, focusing on high-confidence samples, cluster-based groups, and difficult boundary cases.
- The process followed an **iterative workflow**:
- Train → Analyse embeddings → Label useful samples → Retrain.

- With each iteration, more labelled samples were added to the training dataset, improving the dataset quality and model performance.
- Finally, the trained model was used to generate predictions for the **test dataset**, and the results were exported as a **submission.csv file** for evaluation on the Kaggle leaderboard.

CONCLUSION:

In this project, a binary image classifier was developed to distinguish between **chihuahuas and muffins** using a **data-centric AI approach**. Instead of modifying the model architecture, the focus was on improving the quality of the training dataset.

Using the **3LC dashboard**, embeddings and model predictions were analysed to identify useful unlabelled samples. These samples were strategically labelled and added to the training dataset through multiple iterations. This iterative process helped improve the dataset quality and enhanced the model's ability to correctly classify visually similar images.

The results demonstrate that **improving the dataset through strategic labelling can significantly enhance model performance**, even when the model architecture remains unchanged. This highlights the importance of data-centric methods in modern machine learning workflows.